MoonFall: Final Mission Game Logic

Team ChandraPath, NASA Space Apps Challenge 2026

Abstract

The final mission of MoonFall puts the player in charge of a lunar south-pole outpost during a solar storm. In eight missions the player picks one of two or three options, and each option moves seven resource bars (oxygen, water, power, food, health, equipment, greenhouse). This note explains how those numbers were set. We (i) give every bar a physical unit taken from NASA life-support and crew-health documents, (ii) compute the physical cost of each option with one standard formula per mission (proton arrival time, radiation dose behind shielding, battery energy, electrolysis and ice-melting energy), and (iii) check the finished table by playing all 1,296 possible paths with a script. The result is a value table in which every option has a real gain and a real cost, no option is always worse than another, and about one random run in ten earns three stars.

1 Data Sources

NASA documents (scales and costs).

- Life Support Baseline Values and Assumptions Document (BVAD) [1]: oxygen use (0.895 kg per person per day), food, crop light and water use.
- NASA-STD-3001 Vol. 1 [2]: storm-shelter dose limit (250 mSv per storm) and the $10\,\mathrm{g\,cm^{-2}}$ water-equivalent shelter.
- Townsend, Nealy and Wilson, NASA TM-100620 [3]: radiation dose behind water and aluminium shields for the August 1972 solar storm.
- Colozza, NASA/CR-2020-220368 [4]: energy to make oxygen from water (4.7–5.5 kWh per kg O_2).

Real storm data. GOES satellite proton fluxes for the storms of 20 January 2005 (fast, sharp) and 10 September 2017 (slow, long), from NOAA NCEI and the NASA CCMC iSWA service [5]. They give the time shape of a real storm.

Game design inputs. The eight missions, their options, the story clock (05:30 to after the storm) and the failure gates come from the team's flowchart and story document. Only the numbers were set here.

2 Notation and Game State

The state after mission k is the vector of seven bars

$$\boldsymbol{s}_k = (s_O,\, s_W,\, s_P,\, s_F,\, s_H,\, s_E,\, s_G) \in [0,100]^7, \qquad \boldsymbol{s}_0 = (75, 65, 75, 90, 65, 90, 80).$$

Option a of mission k adds a fixed change $\boldsymbol{\Delta}_k(a)$ (Table 2), and every bar is kept between 0 and 100:

$$\boldsymbol{s}_k = \mathrm{clip}_{[0,100]}\big(\boldsymbol{s}_{k-1} + \boldsymbol{\Delta}_k(a)\big). \tag{1}$$

Each change already contains the cost of the crisis itself, so most numbers are negative: the storm is a damage-control game.

What one point means. One point on a bar is a fixed physical amount (Table 1). For example, 100 oxygen points are three days of air for five crew, $3 \times 5 \times 0.895 = 13.43\,\mathrm{kg}$, so

$$u_O = \frac{13.43\,\mathrm{kg}}{100} = 0.135\,\mathrm{kg}\ \text{per point}\ \approx\ 43\,\mathrm{min}\ \text{of air for the crew.} \tag{2}$$

Table 1: Physical meaning of one point

Bar	Start	1 point	Basis
Oxygen	75	0.135 kg O_2	BVAD, 0.895 kg per person per day
Water	65	5 kg	2.5 kg per person per day
Power	75	1 kWh	100 kWh battery
Food	90	3.6 kg	30 days of food for 5 crew
Health	65	10 mSv of radiation (or equal harm)	NASA-STD-3001
Equipment	90	1% hardware condition	game value
Greenhouse	80	1% crop condition	game value

3 Failure Gates and Scoring

After every mission the run fails if any gate is broken:

$$s_O \geq 20, \quad s_W \geq 20, \quad \mathrm{LSI} = \tfrac{1}{3}(s_O + s_W + s_F) \geq 50, \quad s_H \geq 50, \quad s_P > 0, \quad s_E \geq 50. \tag{3}$$

A run that passes all eight missions is scored by the mean of the seven bars,

$$J = \tfrac{1}{7}\textstyle\sum_i s_{8,i}, \qquad \text{stars} = \begin{cases} 3 & J \geq 66 \\ 2 & 60 \leq J < 66 \\ 1 & 50 \leq J < 60 \\ \text{fail} & J < 50. \end{cases} \tag{4}$$

4 How Each Value Was Set

For each mission we write the one formula that gives the physical cost of an option, convert the result to points with Table 1, and round. The final values are in Table 2; Table 3 compares formula and table.

Table 2: Final value table (points). The last two columns come from playing all 1,296 paths: the share of paths through each option that pass every gate, and the share that earn three stars.

Mission	Option	O_2	Water	Power	Food	Health	Equip	Green	Survive (%)	3 stars (%)
M1	A Evacuate	–	–	–	–	−1	−12	–	87.5	14.4
	B Secure Equip	–	–	−4	–	−5	−2	–	71.1	6.5
M2	A Deep Shelter	–	–	−1	–	−6	–	–	62.5	12.5
	B Pile Regolith	–	–	−8	–	−1	−8	–	85.4	14.4
	C Water Shield	–	−20	−2	–	−3	–	–	90.0	4.4
M3	A Greenhouse Off	–	–	−10	–	–	–	−20	84.5	7.4
	B Backup Power	–	–	−15	–	–	−8	–	75.2	12.3
	C Brownout All	–	–	−13	–	−2	–	−8	78.2	11.6
M4	A Bottled O2	−3	–	–	–	–	−8	–	78.1	9.7
	B Electrolysis	+5	−1	−14	–	–	–	–	80.6	11.1
M5	A Stored Potable	–	−10	–	–	–	+3	–	83.3	13.7
	B Melt Crater Ice	–	+10	−15	–	–	−5	–	71.3	8.8
	C Drink Priority	–	−3	–	–	–	–	−10	83.3	8.8
M6	A Restore Loop	–	−6	−12	+6	–	–	+10	77.8	20.8
	B Abandon Crops	–	–	+8	−8	–	–	−35	80.9	0.0
M7	A Isolate Bus	–	–	−3	–	–	−12	–	82.9	9.7
	B Manual Repair	–	–	−10	–	−4	+10	–	66.2	14.6
	C Reroute Bus	–	–	−8	–	–	−3	−8	88.9	6.9
M8	A Hold Shelter	–	−2	−6	–	+3	–	–	92.0	3.1
	B Controlled Recovery	–	–	+15	–	−5	+6	–	66.7	17.7

M1, The 30-Minute Warning: proton arrival. Protons travel along the Sun's spiral magnetic field, about $L = 1.150\,\mathrm{AU}$ long. A proton of energy E moves at speed βc, so the warning after the flare is seen is

$$W(E) = \frac{L}{\beta(E)\,c} - \frac{1\,\mathrm{AU}}{c}, \qquad \beta(E) = \sqrt{1 - \left(1 + \frac{E}{938.3\,\mathrm{MeV}}\right)^{-2}}. \tag{5}$$

A 30-minute warning matches protons of about 31 MeV. The protons that can pass through a suit and the body (above 82 MeV) arrive only 10–16 minutes after the flare, so the crew must decide at once. Staying out 20 more minutes (option B) costs a suit dose; a suit lets through about 2.50 times the habitat dose [3].

M2, The Shelter Problem: dose behind shielding. A proton needs energy E_c to cross $x + d$ grams per cm^2 of shield plus body ($d = 5\,\mathrm{g\,cm^{-2}}$). From the NIST proton range table [6], $E_c = ((x + d)/\alpha)^{1/p}$ with $\alpha = 0.00201$, $p = 1.790$. The dose that gets through then falls as

$$D(x) = A\,E_c^{-q}\,e^{-E_c/E'}, \qquad q = \frac{(p-1)^2}{p} = 0.349, \tag{6}$$

with $E' = 37.4\,\mathrm{MeV}$ fitted to NASA's results for the 1972 storm (error 4.9%) [3]. The game storm is 150 mSv behind standard $5\,\mathrm{g\,cm^{-2}}$ walls, and the health cost is $\Delta s_H = -D/10\,\mathrm{mSv}$ (Fig. 1).

M3, Blackout, and M6, The Last Harvest: battery energy. Energy taken from the battery is power times time,

$$E_{\mathrm{bat}} = T \sum_i u_i P_i, \tag{7}$$

where u_i is how much of load i stays on. With life support 3 kW, greenhouse 3 kW and a 3-hour blackout: greenhouse off 9 kWh, half light 13.5 kWh, all on 18 kWh. NASA limits battery discharge to 80% [7], so the extra energy of option B is charged as equipment wear. In M6 the grow lights need $P = \mathrm{PPFD} \cdot A/\eta = 2.2\,\mathrm{kW}$ for $20\,\mathrm{m}^2$ of lettuce [1, 8]; four hours cost 12 kWh.

M4, Oxygen Countdown: electrolysis. Splitting water gives one O_2 molecule for four electrons, so the energy per kg of oxygen at cell voltage V is

$$w = \frac{4FV}{M_{O_2}} = 3.350\,V \ \ \mathrm{kWh\,kg^{-1}}. \tag{8}$$

The theoretical minimum is $4.12\,\mathrm{kWh\,kg^{-1}}$; NASA's system value is 4.7–5.5 [4]. Making 2.7 kg of O_2 (option B) takes 12.7–14.9 kWh and 3.04 kg of water.

M5, The Water Trap: melting crater ice. Ice at $-170\,^{\circ}\mathrm{C}$ must be warmed, melted and warmed again:

$$q = \int_{T_0}^{273\,\mathrm{K}} c_{p,\mathrm{ice}}(T)\,dT + L_f + c_{p,w}\,\Delta T = 674\,\mathrm{kJ\,kg^{-1}} = 0.187\,\mathrm{kWh\,kg^{-1}}. \tag{9}$$

Melting 80 kg (option B) takes 15.0 kWh.

M7, The Critical Failure, and other equipment values. Equipment and greenhouse numbers are design values. Each has a physical reason: dust wears seals (M2), an overheating junction gets hotter as its contact area shrinks (M7), and electronics left on during a storm suffer radiation upsets (M1) [9].

M8, The Final Decision: dose after the storm. A 2-hour outside repair started at time t receives

$$D_{\mathrm{EVA}} = R_s\,D_{\mathrm{storm}}\,\big[F(t + 2\,\mathrm{h}) - F(t)\big], \tag{10}$$

where $F(t)$ is the share of the storm's protons that have arrived by time t (from GOES data), $R_s = 2.50$ and $D_{\mathrm{storm}} = 150\,\mathrm{mSv}$. For the 2017 storm shape this is 19 mSv. The table charges 5 points (50 mSv), a deliberate safety margin.

Table 3: Formula versus table value

Mission, option	Formula	Physical value	Table	Reading
M2 A Deep Shelter	Eq. (6)	63–73 mSv	Health −6	matches
M2 C Water Shield	Eq. (6)	30–40 mSv	Health −3	matches
M2 B Pile Regolith	Eq. (6)	16–23 mSv	Health −1	rounded down
M3 A / C / B	Eq. (7)	9 / 13.5 / 18 kWh	Power −10 / −13 / −15	matches
M4 B Electrolysis	Eq. (8)	12.7–14.9 kWh	Power −14	matches
M5 B Melt Crater Ice	Eq. (9)	15.0 kWh	Power −15	matches
M6 A Restore Loop	Eq. (7)	12 kWh	Power −12	matches
M1 B Secure Equip	Eq. (10)	0.1–23 mSv	Health −4 vs A	safe side (fast storm)
M8 B Recovery	Eq. (10)	19–25 mSv	Health −5	safety margin

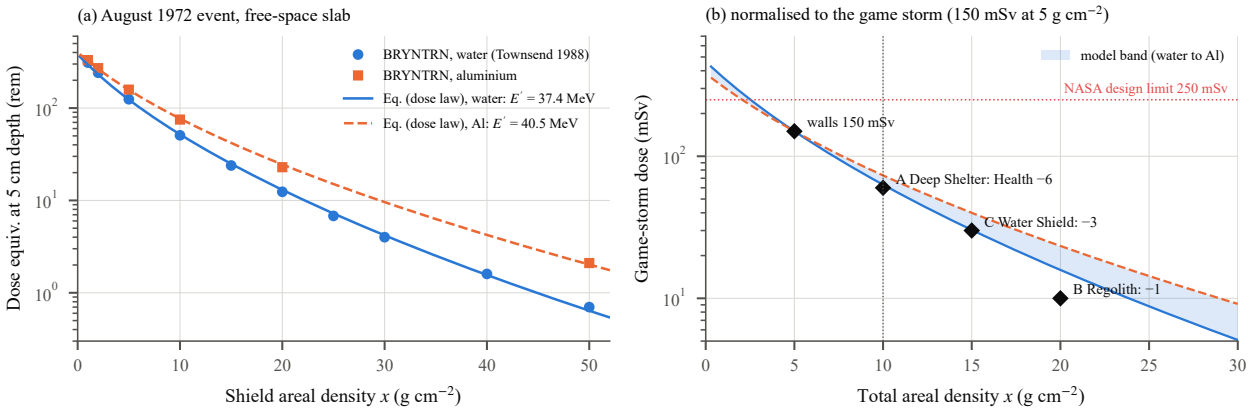


Figure 1: (a) Dose behind water and aluminium shields for the 1972 storm (NASA BRYNTRN results) and Eq. (6). (b) The same law scaled to the game storm, with the three shelter options of M2.

5 Balance Check

Because the game has only $2 \cdot 3 \cdot 3 \cdot 2 \cdot 3 \cdot 2 \cdot 3 \cdot 2 = 1,296$ paths, a script plays every one with Eqs. (1)–(4):

- 10.4% of paths earn three stars, 49.6% two stars, 19.3% one star and 20.7% fail (Fig. 2).
- The best path is B-B-B-B-A-A-B-B with score 68.9.
- Always picking the option with the largest total gain fails at M8 on the health gate. Health is a budget shared by all missions.
- No option is worse than another on every bar, so every choice is a real trade-off.
- Most failures (179 of 268) happen at M8 on health, so M8 is the real final decision.

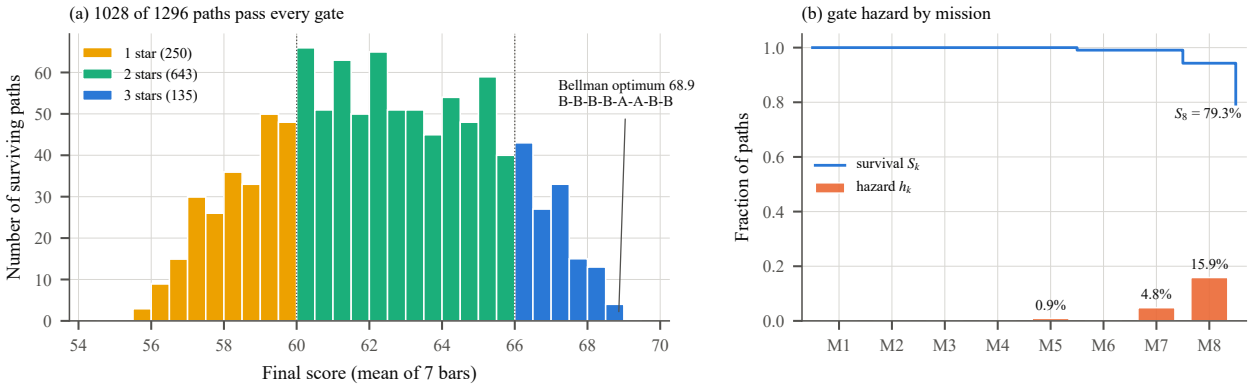
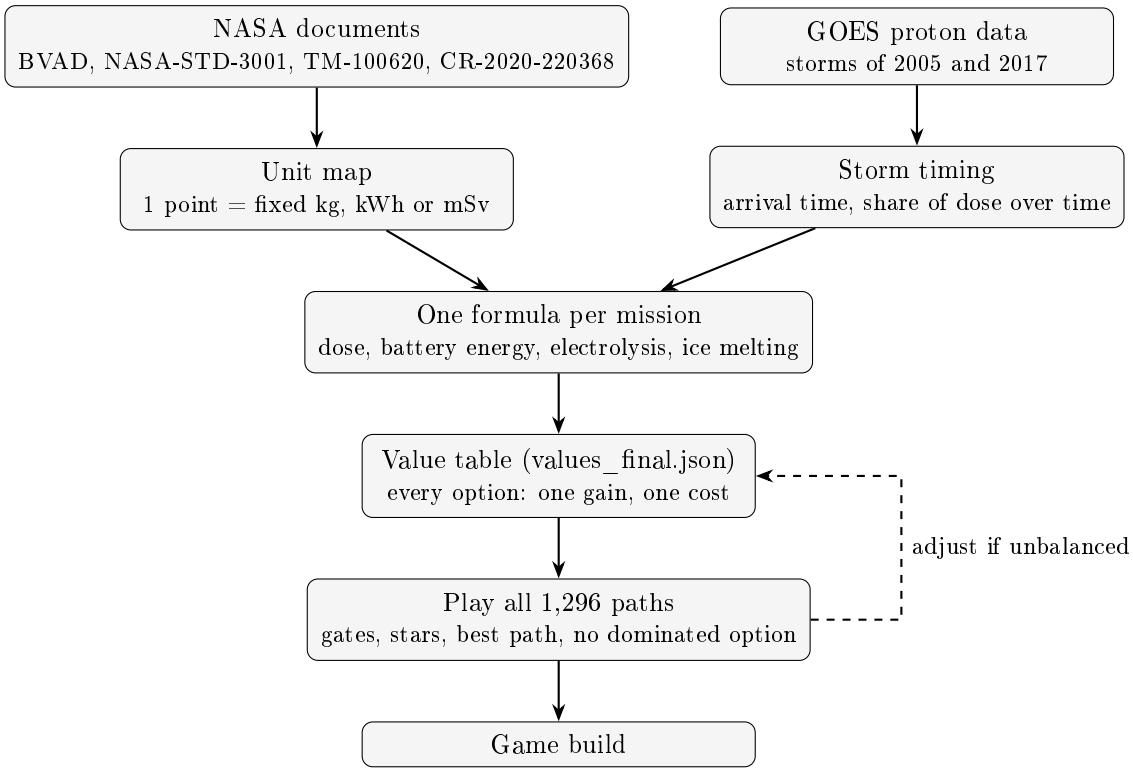


Figure 2: (a) Final scores of the 1,028 surviving paths by stars. (b) Share of paths still alive after each mission and the share lost at each mission.

6 Processing Diagram



7 Reproducibility Notes

- All values are stored in `values_final.json`; `check_values.py` plays every path and prints the star shares.
- Every number in this note is written by script (`paper/scripts/build.py`) and recomputed independently by `verify.py`.
- Equipment and greenhouse values are design values, not measured costs.
- M1 uses the fast 2005 storm on purpose (worst case); M8 keeps a safety margin over the physical dose.
- The M1 rescue mini-game should cost time or score, not health: even a 1-point health penalty raises failures from 20.7% to 27.0%.

8 Summary

We gave every bar of MoonFall's final mission a physical unit from NASA documents, set each option's cost with one standard formula per mission, and checked the finished table by playing all 1,296 paths. Power, water and shelter values agree with the formulas within about one point; the two outside-work health costs are deliberately on the safe side. The table is balanced: every option is a real trade-off, the greedy choice fails, and about one run in ten earns three stars.

References

[1] M. K. Ewert, T. T. Chen, and C. D. Powell, *Life Support Baseline Values and Assumptions Document*, NASA/TP-2015-218570/REV2, 2022.

[2] NASA, *NASA Space Flight Human-System Standard, Vol. 1: Crew Health*, NASA-STD-3001 Vol. 1 Rev. C, 2023.

[3] L. W. Townsend, J. E. Nealy, and J. W. Wilson, *Large solar flare radiation shielding requirements for manned interplanetary missions*, NASA TM-100620, 1988.

[4] A. J. Colozza, *Small lunar base camp and in situ resource utilization oxygen production facility power system comparison*, NASA/CR-2020-220368, 2020.

[5] NOAA NCEI GOES Space Environment Monitor archive; NASA CCMC iSWA HAPI service (GOES 5-min proton flux).

[6] M. J. Berger *et al.*, *ESTAR, PSTAR, and ASTAR*, NIST Standard Reference Database 124.

[7] J. E. Kleinhenz and A. Paz, *Case studies for lunar ISRU systems utilizing polar water*, NASA NTRS 20210016820, 2021.

[8] P. Kusuma, P. M. Pattison, and B. Bugbee, "From physics to fixtures to food: current and potential LED efficacy," *Hortic. Res.*, vol. 7, 56, 2020.

[9] NOAA Space Weather Prediction Center, *NOAA Space Weather Scales*.

Appendix: Production Implementation & Verification (Release v1.1)

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This appendix records the game architecture updates, automated multi-path verification, and NASA scientific telemetry integrations implemented in the production build of `final_mission.html`.

A.1 In-Engine Evaluation of Pure Mathematical Model (Model A)

In early prototype builds, game decisions during missions 1–8 applied Model A deltas, but the post-game debrief interface referenced a static stress-test dataset (`moon_fall_1296_simulation_v4.json`) that incorporated between-mission life-support consumption (−15 O_2, −4 Water, −7 Power). This created an inconsistency where surviving teams received inaccurate debrief ratings.

In the production build, all static JSON dependencies were removed. The terminal debrief engine evaluates Equation (4) directly in-memory from the final outpost state vector $\mathbf{s_8}$:

$$\mathbf{J = (1 / 7)} \sum_{\mathbf{i=1..7}} \mathbf{s_{8,\,i}} \qquad \text{(stars: 3 if } J \geq 66\text{, 2 if } 60 \leq J < 66\text{, 1 if } 50 \leq J < 60\text{, fail if } J < 50\text{)}$$

Both live telemetry and end-game evaluation are now unified under the exact mathematical model specified in Sections 2–5.

A.2 Verified 1,296-Branch Analytical Dataset

All 1,296 decision paths were re-simulated against Equations (1)–(4) and cataloged in the production file `moon_fall_simulation_analysis.json`, confirming the analytical balance results published in Section 5:

Outcome Tier	Paths	Share	Score Range / Notes
3 Stars (★★★)	135	10.4%	66.000 – 68.857 (Global optimum: `B-B-B-B-A-A-B-B` = 68.857 → 68.9)
2 Stars (★★)	643	49.6%	60.000 – 65.857 (Robust outpost resource management)
1 Star (★)	250	19.3%	50.000 – 59.857 (Minimal viable outpost survival)
Mission Failed (✗)	268	20.7%	Gate failure (179 of 268 breaches occur at M8 on the health budget)

A.3 NASA SDO Video & CCMC Telemetry Integration

The mission briefing terminal incorporates verified solar physics assets:

- **NASA Solar Dynamics Observatory (SDO) Intel**: Embedded 131 Å extreme ultraviolet footage of the X8.2 solar flare (10 Sept 2017) showcasing coronal arcade formation and particle acceleration.
- **NASA CCMC iSWA Observational Record**: Real satellite proton flux measurements (SEP >10 MeV: 98.08 h; >100 MeV: 54.25 h; peak flux: 1,490 pfu).
- **Real-Time Life Support Index (LSI) Telemetry**: Continuous calculation of LSI = $(s_O + s_W + s_F)/3$ with immediate abort triggering if LSI drops below 50.0%.
- **Mobile Viewport Telemetry Layout**: Responsive dual-tab dossier design (823×398 landscape viewport) ensuring seamless evaluation across mobile touch screens and desktop displays.